1 Strang Splitting for the Time-Dependent Schrödinger Equation Using FFT

1.1 Introduction

The time-dependent Schrödinger equation (TDSE) for a nonrelativistic quantum particle in one spatial dimension is

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = \hat{H}\psi(x,t), \tag{1}$$

where the Hamiltonian is usually written as a sum of kinetic and potential contributions,

$$\hat{H} = \hat{T} + \hat{V}. \tag{2}$$

For a particle of mass m moving in a scalar potential $V(x,t)$,

$$\hat{T} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2}, \qquad \hat{V} = V(x,t). \tag{3}$$

The formal solution of the TDSE from time t to time $t+\Delta t$ is

$$\psi(x,t+\Delta t) = \hat{U}(t+\Delta t,t)\psi(x,t), \tag{4}$$

where the evolution operator is

$$\hat{U}(t+\Delta t,t) = \exp\left[-\frac{i}{\hbar}\hat{H}\Delta t\right] \tag{5}$$

when the Hamiltonian is time-independent over the interval. Even for time-dependent Hamiltonians, one often approximates the propagator over a sufficiently small step Δt by freezing the Hamiltonian appropriately within the interval.

The central numerical difficulty is that $\hat{T}$ and $\hat{V}$ generally do not commute:

$$[\hat{T},\hat{V}] \neq 0. \tag{6}$$

Therefore,

$$e^{-\frac{i}{\hbar}(\hat{T}+\hat{V})\Delta t} \neq e^{-\frac{i}{\hbar}\hat{T}\Delta t}e^{-\frac{i}{\hbar}\hat{V}\Delta t}. \tag{7}$$

The split-operator method, more specifically the *Strang splitting*, provides a highly efficient and physically faithful approximation to this evolution. When combined with the Fast Fourier Transform (FFT), it becomes one of the most widely used methods for simulating wavepacket dynamics.

1.2 Basic Idea of Operator Splitting

Suppose we want to approximate the exponential of a sum of two noncommuting operators, $\hat{A}$ and $\hat{B}$. A first-order splitting is

$$e^{(\hat{A}+\hat{B})\Delta t} = e^{\hat{A}\Delta t}e^{\hat{B}\Delta t} + O(\Delta t^2), \tag{8}$$

but this is only first-order accurate globally in time.

A much better approximation is the *Strang splitting*,

$$e^{(\hat{A}+\hat{B})\Delta t} = e^{\hat{A}\Delta t/2}e^{\hat{B}\Delta t}e^{\hat{A}\Delta t/2} + O(\Delta t^3). \tag{9}$$

This local error of order $O(\Delta t^3)$ yields a second-order accurate method globally over a fixed total propagation time.

For the TDSE, we choose

$$\hat{A} = -\frac{i}{\hbar}\hat{V}, \qquad \hat{B} = -\frac{i}{\hbar}\hat{T}, \tag{10}$$

so that the short-time propagator is approximated by

$$e^{-\frac{i}{\hbar}(\hat{T}+\hat{V})\Delta t} \approx e^{-\frac{i}{2\hbar}\hat{V}\Delta t} e^{-\frac{i}{\hbar}\hat{T}\Delta t} e^{-\frac{i}{2\hbar}\hat{V}\Delta t}. \tag{11}$$

This is the standard split-operator propagator used in quantum dynamics.

1.3 Why FFT?

The usefulness of the method lies in the observation that the two parts of the Hamiltonian are diagonal in different representations:

- The potential operator $\hat{V} = V(x)$ is diagonal in position space.
- The kinetic operator $\hat{T} = \hat{p}^2/(2m)$ is diagonal in momentum space.

Therefore,

$$e^{-\frac{i}{2\hbar}\hat{V}\Delta t} \tag{12}$$

acts as a simple multiplication in position space,

$$\psi(x) \mapsto e^{-\frac{i}{2\hbar}V(x)\Delta t}\psi(x), \tag{13}$$

while

$$e^{-\frac{i}{\hbar}\hat{T}\Delta t} \tag{14}$$

acts as a simple multiplication in momentum space,

$$\tilde{\psi}(p) \mapsto e^{-\frac{i}{\hbar}\frac{p^2}{2m}\Delta t}\tilde{\psi}(p). \tag{15}$$

Thus, one performs the following sequence at each time step:

1. Apply half of the potential evolution in position space.
2. Fourier transform to momentum space.
3. Apply the kinetic evolution in momentum space.
4. Inverse Fourier transform back to position space.
5. Apply the remaining half of the potential evolution in position space.

The FFT makes the transformations between position and momentum representations computationally efficient, reducing the cost from $O(N^2)$ for a direct discrete Fourier transform to $O(N \log N)$.

1.4 Fourier Representation

Let the momentum-space wavefunction be defined by the Fourier transform

$$\tilde{\psi}(k,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \psi(x,t) e^{-ikx}\, dx, \tag{16}$$

with inverse transform

$$\psi(x,t) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\infty} \tilde{\psi}(k,t) e^{ikx}\, dk. \tag{17}$$

Here $p = \hbar k$, so in k-space the kinetic operator becomes

$$\hat{T} = \frac{\hbar^2 k^2}{2m}. \tag{18}$$

Hence the kinetic propagator is

$$e^{-\frac{i}{\hbar}\hat{T}\Delta t} = e^{-i\frac{\hbar k^2}{2m}\Delta t}. \tag{19}$$

If one works directly with momentum p instead of wave number k, then

$$e^{-\frac{i}{\hbar}\hat{T}\Delta t} = e^{-i\frac{p^2}{2m\hbar}\Delta t}. \tag{20}$$

1.5 Derivation of the Strang-Split Propagator

To see why Eq. (11) is second-order accurate, one may invoke the Baker–Campbell–Hausdorff (BCH) expansion. Let

$$\hat{X} = -\frac{i}{\hbar}\hat{V}\Delta t, \qquad \hat{Y} = -\frac{i}{\hbar}\hat{T}\Delta t. \tag{21}$$

Then Strang splitting corresponds to

$$e^{\hat{X}/2} e^{\hat{Y}} e^{\hat{X}/2}. \tag{22}$$

A BCH analysis shows that

$$e^{\hat{X}/2} e^{\hat{Y}} e^{\hat{X}/2} = \exp\left(\hat{X} + \hat{Y} + \frac{1}{12}[\hat{X},[\hat{X},\hat{Y}]] - \frac{1}{24}[\hat{Y},[\hat{X},\hat{Y}]] + \cdots \right). \tag{23}$$

Since $\hat{X}$ and $\hat{Y}$ are each $O(\Delta t)$, the leading correction terms are $O(\Delta t^3)$. Importantly, the $O(\Delta t^2)$ term cancels because of the symmetry of the composition. This symmetry is also responsible for time reversibility:

$$\hat{U}_{\text{Strang}}(-\Delta t) = \hat{U}_{\text{Strang}}(\Delta t)^{-1}. \tag{24}$$

1.6 Algorithm for the Time-Independent Potential

For a time-independent potential $V(x)$, define

$$U_V^{1/2}(x) = \exp\left[-\frac{i}{2\hbar} V(x)\Delta t \right], \tag{25}$$

and

$$U_T(k) = \exp\left[-i\frac{\hbar k^2}{2m}\Delta t \right]. \tag{26}$$

Then one time step from t_n to $t_{n+1} = t_n + \Delta t$ proceeds as follows:

$$\psi^{(1)}(x) = U_V^{1/2}(x)\psi(x, t_n), \tag{27}$$

$$\tilde{\psi}^{(1)}(k) = \mathcal{F}\left[\psi^{(1)}(x)\right], \tag{28}$$

$$\tilde{\psi}^{(2)}(k) = U_T(k)\tilde{\psi}^{(1)}(k), \tag{29}$$

$$\psi^{(2)}(x) = \mathcal{F}^{-1}\left[\tilde{\psi}^{(2)}(k)\right], \tag{30}$$

$$\psi(x, t_{n+1}) = U_V^{1/2}(x)\psi^{(2)}(x). \tag{31}$$

Equivalently, the full update can be written compactly as

$$\psi(x, t + \Delta t) = U_V^{1/2}(x)\,\mathcal{F}^{-1}\left\{U_T(k)\,\mathcal{F}\left[U_V^{1/2}(x)\psi(x, t)\right]\right\}. \tag{32}$$

1.7 Time-Dependent Potentials

If the potential depends on time, $V(x, t)$, then the exact propagator involves time ordering:

$$\hat{U}(t + \Delta t, t) = \mathcal{T}\exp\left[-\frac{i}{\hbar}\int_t^{t+\Delta t} \hat{H}(t')\,dt'\right]. \tag{33}$$

A common second-order approximation is to evaluate the potential at the midpoint of the interval:

$$\hat{U}(t + \Delta t, t) \approx e^{-\frac{i}{2\hbar}\hat{V}(t+\Delta t/2)\Delta t} e^{-\frac{i}{\hbar}\hat{T}\Delta t} e^{-\frac{i}{2\hbar}\hat{V}(t+\Delta t/2)\Delta t}. \tag{34}$$

This retains second-order accuracy provided the Hamiltonian varies smoothly in time.

More generally, if there is also an explicitly time-dependent vector potential or if the kinetic term itself becomes time-dependent, then extra care is needed, because the straightforward diagonalization of the kinetic part in momentum space may no longer hold in such a simple way.

1.8 Spatial Discretization

Consider a finite spatial interval of length L,

$$x \in [x_{\min}, x_{\max}), \qquad L = x_{\max} - x_{\min}, \tag{35}$$

discretized into N grid points,

$$x_j = x_{\min} + j\Delta x, \qquad j = 0, 1, \ldots, N-1, \qquad \Delta x = \frac{L}{N}. \tag{36}$$

The associated discrete wave-number grid for FFT conventions is

$$k_n = \frac{2\pi}{L}\begin{cases} n, & 0 \le n \le \frac{N}{2} - 1, \\ n - N, & \frac{N}{2} \le n \le N - 1. \end{cases} \tag{37}$$

This arrangement reflects the periodic structure implicit in the FFT.

The discrete kinetic propagator is therefore

$$U_T(k_n) = \exp\left[-i\frac{\hbar k_n^2}{2m}\Delta t\right]. \tag{38}$$

1.9 Periodic Boundary Condition and Its Consequences

The use of FFT implies that the computational domain is effectively treated as periodic:

$$\psi(x + L, t) = \psi(x, t). \tag{39}$$

This is not necessarily a physical periodicity of the original problem; rather, it is a numerical consequence of representing the wavefunction on a finite Fourier grid. Therefore, the domain must be chosen large enough so that the wavefunction remains negligible near the boundaries during the simulation. Otherwise, outgoing wavepackets will re-enter from the opposite boundary, producing unphysical wrap-around effects.

This is particularly important in scattering problems. One must ensure that:

- the initial wavepacket is sufficiently localized away from the boundaries,
- the scattered components do not reach the boundaries before the simulation ends, or
- an absorbing boundary method is added.

1.10 Norm Conservation and Unitarity

For a real time-independent potential and exact FFT operations, the Strang split-operator method is unitary. Each sub-step is unitary:

- Multiplication by
$$e^{-\frac{i}{2\hbar}V(x)\Delta t} \tag{40}$$
has modulus one pointwise.
- Fourier transform is unitary, up to normalization conventions.
- Multiplication by
$$e^{-i\frac{\hbar k^2}{2m}\Delta t} \tag{41}$$
also has modulus one.

Therefore, in exact arithmetic,

$$\int |\psi(x, t)|^2 \, dx \tag{42}$$

is preserved at every time step.

In floating-point arithmetic there will be tiny numerical roundoff errors, but norm preservation is usually extremely good.

1.11 Accuracy and Error Analysis

The method has local error

$$O(\Delta t^3), \tag{43}$$

which leads to global error

$$O(\Delta t^2) \tag{44}$$

over a fixed total evolution time.

The dominant error originates from nested commutators such as

$$[\hat{V}, [\hat{V}, \hat{T}]], \qquad [\hat{T}, [\hat{V}, \hat{T}]]. \tag{45}$$

This is physically reasonable: the more strongly kinetic and potential evolutions fail to commute, the more splitting error one expects.

For smooth potentials and sufficiently small Δt, the method is very accurate. However, one must remember that there are *two* numerical discretizations:

1. time discretization through operator splitting,
2. space discretization through the finite grid and the spectral representation.

Thus convergence must be checked with respect to both Δt and Δx.

1.12 Spectral Accuracy in Space

A major strength of the FFT-based split-operator method is that the kinetic term is treated spectrally. For smooth wavefunctions on a sufficiently fine periodic grid, Fourier methods have excellent spatial accuracy, often far superior to low-order finite-difference approximations. In this sense, the method combines:

- second-order accuracy in time,
- spectral or near-spectral accuracy in space.

This is one reason why it is extremely attractive for wavepacket propagation.

1.13 Comparison with Finite-Difference Methods

It is useful to compare the split-operator FFT method with finite-difference methods such as explicit Euler, Crank–Nicolson, or Runge–Kutta applied to the discretized TDSE.

Advantages of the split-operator FFT method:

- Very efficient: $O(N \log N)$ per time step.
- Naturally unitary for Hermitian Hamiltonians.
- Excellent spatial accuracy for smooth problems.
- Particularly elegant for Hamiltonians of the form $\hat{T} + \hat{V}(x)$.

Limitations:

- Implicitly periodic due to FFT.
- Less convenient for complicated boundary conditions.
- Less straightforward when the kinetic operator is not diagonal in Fourier space.
- Care is needed for nonsmooth potentials or sharply localized singular structures.

Compared with Crank–Nicolson: Crank–Nicolson is also unitary for Hermitian Hamiltonians and second-order accurate in time, but it requires solving a linear system at each time step. For simple one-dimensional problems this is manageable, but in higher dimensions or for repeated long-time propagation, the FFT-based split operator can be substantially faster.

1.14 Extension to Higher Dimensions

In two or three dimensions, the method generalizes directly. For example, in three dimensions,

$$\hat{T} = -\frac{\hbar^2}{2m}\nabla^2 = \frac{\hat{p}_x^2 + \hat{p}_y^2 + \hat{p}_z^2}{2m}. \tag{46}$$

In momentum space this is diagonal:

$$T(\mathbf{k}) = \frac{\hbar^2|\mathbf{k}|^2}{2m}. \tag{47}$$

Hence the kinetic propagator becomes

$$U_T(\mathbf{k}) = \exp\left[-i\frac{\hbar|\mathbf{k}|^2}{2m}\Delta t\right]. \tag{48}$$

The rest of the Strang splitting is unchanged, except that one uses multidimensional FFTs.

Thus in d dimensions,

$$\psi(\mathbf{r}, t+\Delta t) = U_V^{1/2}(\mathbf{r})\,\mathcal{F}^{-1}\left[U_T(\mathbf{k})\,\mathcal{F}\left(U_V^{1/2}(\mathbf{r})\psi(\mathbf{r},t)\right)\right]. \tag{49}$$

1.15 Worked-Out Formula for a One-Dimensional Particle

For clarity, let us write the one-dimensional scheme explicitly. Suppose

$$i\hbar\frac{\partial\psi(x,t)}{\partial t} = \left[-\frac{\hbar^2}{2m}\frac{\partial^2}{\partial x^2} + V(x)\right]\psi(x,t). \tag{50}$$

Then over one time step,

$$\psi(x, t+\Delta t) \approx e^{-\frac{i}{2\hbar}V(x)\Delta t}\,\mathcal{F}^{-1}\left[e^{-i\frac{\hbar k^2}{2m}\Delta t}\,\mathcal{F}\left(e^{-\frac{i}{2\hbar}V(x)\Delta t}\psi(x,t)\right)\right]. \tag{51}$$

If one sets $\hbar = 1$ and $m = 1$, then this simplifies to

$$\psi(x, t+\Delta t) \approx e^{-iV(x)\Delta t/2}\,\mathcal{F}^{-1}\left[e^{-ik^2\Delta t/2}\,\mathcal{F}\left(e^{-iV(x)\Delta t/2}\psi(x,t)\right)\right]. \tag{52}$$

This is the form frequently implemented in teaching codes and research scripts.

1.16 Choice of Time Step and Grid Spacing

The time step Δt should be chosen small enough that the local splitting error remains negligible. Roughly speaking, faster phase oscillations require smaller Δt. The highest resolved kinetic phase on the grid is associated with the largest wave number,

$$k_{\max} \sim \frac{\pi}{\Delta x}. \tag{53}$$

Hence the fastest kinetic phase rotation per step is of order

$$\frac{\hbar k_{\max}^2}{2m}\Delta t. \tag{54}$$

If this becomes too large, the simulation may still remain formally stable, but accuracy will degrade.

Likewise, the grid spacing Δx must resolve:

- the initial wavepacket width,

- any fine structure that develops in the wavefunction,
- the relevant de Broglie wavelengths,
- rapid spatial variations in the potential.

A practical convergence test is essential:

1. run with $(\Delta x, \Delta t)$,
2. refine to $(\Delta x/2, \Delta t/2)$,
3. compare observables such as norm, energy expectation, transmission probability, or overlap fidelity.

1.17 Observables During Propagation

Once $\psi(x,t)$ is known, one can compute observables at each step:

$$\langle x \rangle = \int \psi^*(x,t)\, x\, \psi(x,t)\, dx, \tag{55}$$

$$\langle p \rangle = \int \tilde{\psi}^*(k,t)\, \hbar k\, \tilde{\psi}(k,t)\, dk, \tag{56}$$

$$\langle H \rangle = \langle \psi | \hat{T} + \hat{V} | \psi \rangle. \tag{57}$$

For time-independent Hamiltonians, $\langle H \rangle$ should remain nearly constant. Small deviations provide a useful diagnostic of numerical accuracy.

1.18 Typical Application: Gaussian Wavepacket Scattering

A standard example is the scattering of a Gaussian wavepacket,

$$\psi(x,0) = \frac{1}{(2\pi\sigma^2)^{1/4}} \exp\left[-\frac{(x-x_0)^2}{4\sigma^2} + ik_0 x \right], \tag{58}$$

from a localized potential barrier or well. The split-operator FFT method is especially suitable here because:

- the initial state is smooth,
- the kinetic step is exact in Fourier space,
- long-time propagation is efficient,
- reflection and transmission can be read off from the wavepacket components.

One can define the transmission probability at late times as

$$T = \int_{x > x_b} |\psi(x, t_{\text{final}})|^2\, dx, \tag{59}$$

where x_b lies sufficiently far to the right of the barrier region, once the transmitted packet has separated.

1.19 Imaginary-Time Propagation

An interesting variation is obtained by replacing real time with imaginary time:

$$t \to -i\tau. \tag{60}$$

Then the TDSE becomes a diffusion-like equation,

$$-\hbar \frac{\partial \psi}{\partial \tau} = \hat{H}\psi, \tag{61}$$

whose formal solution is

$$\psi(\tau) = e^{-\hat{H}\tau/\hbar}\psi(0). \tag{62}$$

Repeated propagation damps excited-state components faster than the ground state, allowing one to approximate the ground-state wavefunction after repeated normalization.

The Strang split form becomes

$$e^{-(\hat{T}+\hat{V})\Delta\tau/\hbar} \approx e^{-\hat{V}\Delta\tau/2\hbar} e^{-\hat{T}\Delta\tau/\hbar} e^{-\hat{V}\Delta\tau/2\hbar}. \tag{63}$$

Here the sub-operators are no longer unitary, but the FFT strategy remains equally effective.

1.20 Higher-Order Splittings

Strang splitting is second-order accurate. If higher accuracy is needed, one may compose Strang steps in carefully chosen ways to obtain fourth-order or even higher-order methods. However, these typically require more substeps and may involve negative time coefficients. In many practical TDSE simulations, Strang splitting offers an excellent balance between simplicity, robustness, and efficiency.

1.21 Practical Merits of the Method

The FFT-based Strang split method is popular because it satisfies several highly desirable properties simultaneously:

1. It respects the structure of the Hamiltonian decomposition.
2. It is explicit in implementation.
3. It preserves norm for Hermitian dynamics.
4. It is computationally efficient.
5. It is conceptually transparent and easy to teach.
6. It works naturally for wavepacket propagation, tunneling, scattering, and coherent control problems.

For these reasons it is often the first spectral propagator introduced in computational quantum mechanics.

1.22 Summary

The TDSE with Hamiltonian $\hat{H} = \hat{T} + \hat{V}$ can be efficiently solved numerically by combining Strang splitting with FFT. The central approximation is

$$e^{-\frac{i}{\hbar}(\hat{T}+\hat{V})\Delta t} \approx e^{-\frac{i}{2\hbar}\hat{V}\Delta t} e^{-\frac{i}{\hbar}\hat{T}\Delta t} e^{-\frac{i}{2\hbar}\hat{V}\Delta t}, \tag{64}$$

which is second-order accurate in time. Since $\hat{V}$ is diagonal in position space and $\hat{T}$ is diagonal in momentum space, the two exponentials can each be applied by simple pointwise multiplication, with FFT and inverse FFT used to move between representations.

Thus one time step consists of:

1. half potential kick in position space,
2. FFT to momentum space,
3. full kinetic evolution in momentum space,
4. inverse FFT back to position space,
5. half potential kick in position space.

This yields a method that is efficient, unitary, time-reversible, and spectrally accurate in space for smooth problems. Its main caveat is the periodicity inherent in the FFT grid, which must be handled through sufficiently large domains or absorbing boundaries. Within those constraints, the FFT-based Strang split operator method is among the most elegant and useful tools for numerical quantum dynamics.

1.23 Compact Algorithmic Form

For ready reference, the update rule may be written as

$$\boxed{\psi(x, t + \Delta t) = e^{-\frac{i}{2\hbar}V(x)\Delta t}\, \mathcal{F}^{-1}\left[e^{-i\frac{\hbar k^2}{2m}\Delta t}\, \mathcal{F}\left(e^{-\frac{i}{2\hbar}V(x)\Delta t}\psi(x,t)\right)\right].} \tag{65}$$

This boxed equation is the essence of the FFT-based Strang split-operator method for the TDSE.